

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T33 · Graphs of Functions

With Answers

Questions with full worked solutions

■ 47 pages

FAMILY MAP

Graphs of Functions

F01	Generate a table and draw a nonlinear graph	8 Q 25 marks
F02	Recognise a function graph family	4 Q 9 marks
F03	Sketch key features without a full table	2 Q 7 marks
F04	Solve using graph roots or intersections	5 Q 17 marks

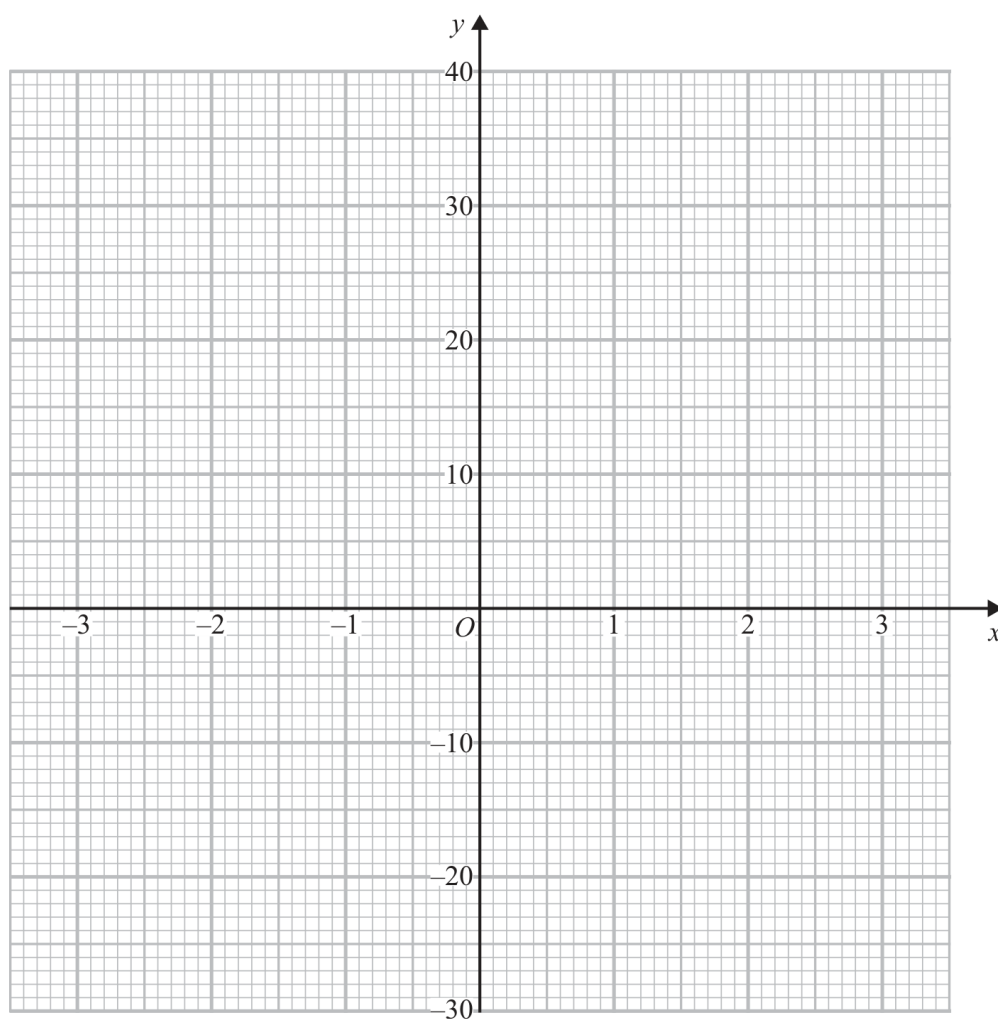
Working Unit 07

4MA1-1H Q10

ORIGINAL QUESTION**3 marks**10 (a) Complete the table of values for $y = x^3 + 2x + 3$

x	-3	-2	-1	0	1	2	3
y		-9	0	3	6		36

(1)

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leq x \leq 3$ 

(2)

Worked Solution 07

Complete and draw a cubic graph

UNIT 1
3 marks**Complete the table**

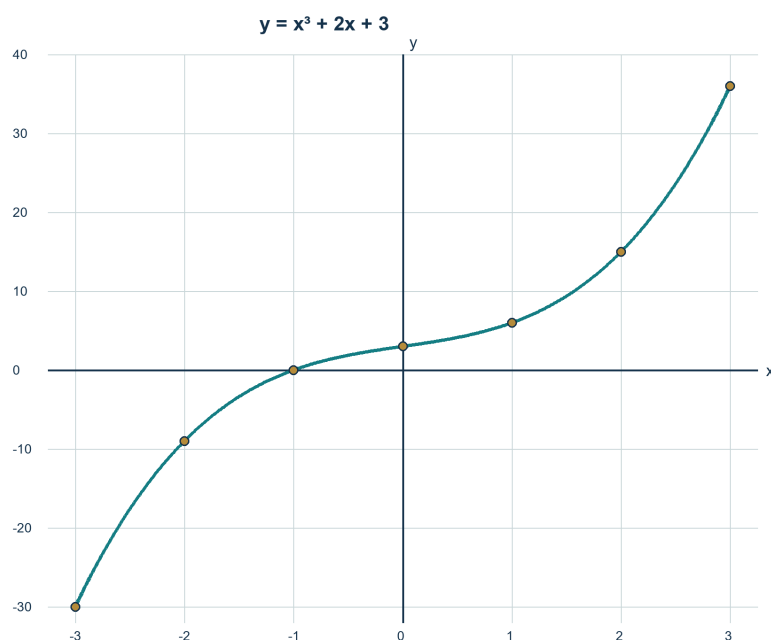
$$y(-3) = -27 - 6 + 3 = -30, \quad y(2) = 8 + 4 + 3 = 15.$$

Plot

$$(-3, -30), (-2, -9), (-1, 0), (0, 3), (1, 6), (2, 15), (3, 36).$$

Join

Draw one smooth increasing cubic curve through all seven points for $-3 \leq x \leq 3$.



Solved reference diagram

Final answer

Missing values -30 and 15 ; smooth cubic through all seven points.

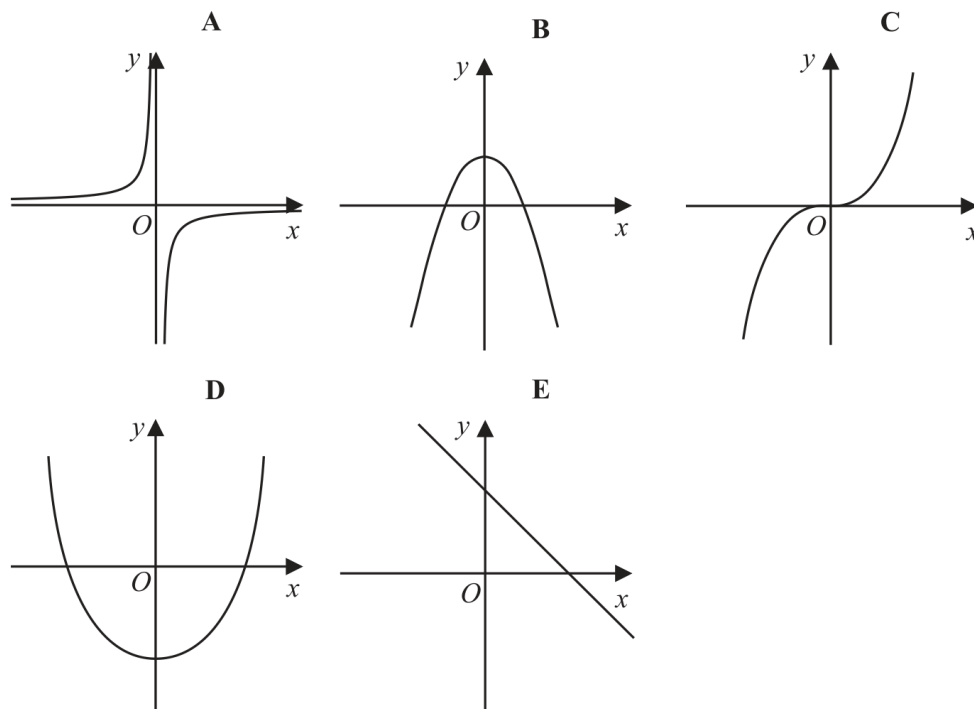
Dependency check The graph uses the completed table; do not use straight-line joins.

Working Unit 20

4MA1-2H Q14

ORIGINAL QUESTION**2 marks**

14 Here are five graphs.



(a) Write down the letter of the graph that could have the equation $y = 5 - x^2$

.....
(1)

(b) Write down the letter of the graph that could have the equation $y = 2x^3$

.....
(1)

Worked Solution 20

Recognise graphs from equations

UNIT 1
2 marks**Part (a)**

$y = 5 - x^2$ is a downward parabola with positive y -intercept: graph B.

Part (b)

$y = 2x^3$ is an increasing cubic through the origin: graph C.

Final answer

(a) B (b) C

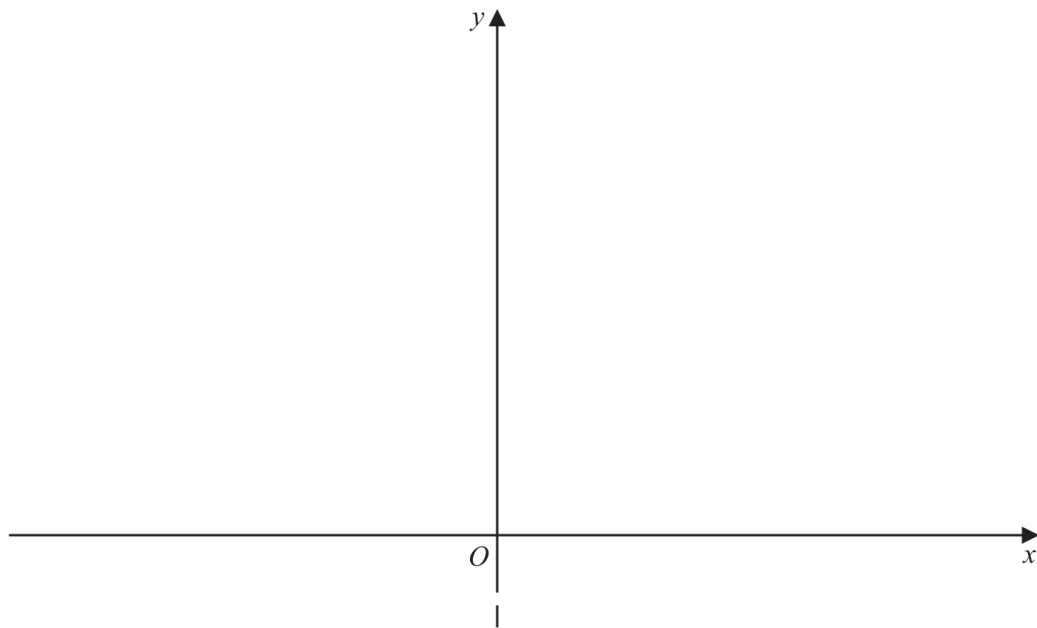
Working Unit 28

4MA1-2H Q25

ORIGINAL QUESTION**3 marks**

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis.



(3)

Worked Solution 28

Sketch a quadratic from completed-square form

UNIT 1
3 marks**Turning point**

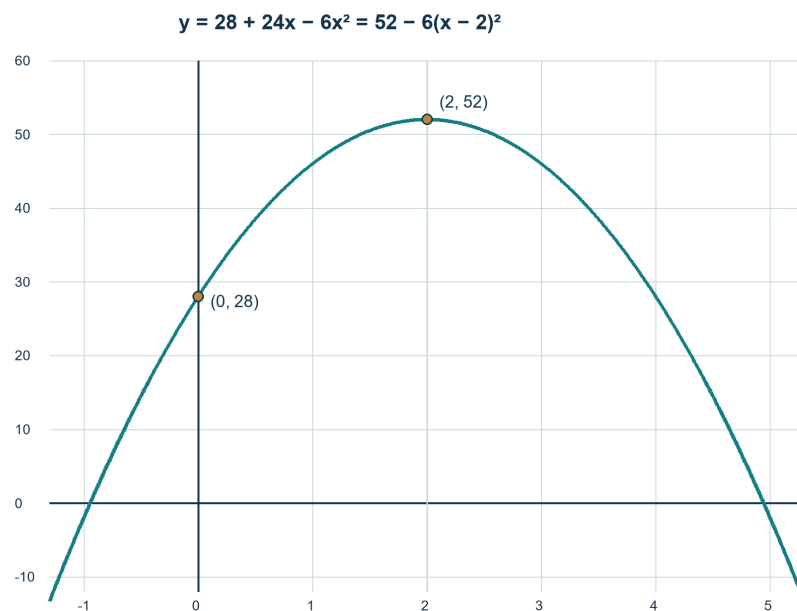
$$y = 52 - 6(x - 2)^2 \Rightarrow \text{maximum turning point } (2, 52).$$

y-intercept

$$x = 0 \Rightarrow y = 28, \text{ so the graph crosses the } y\text{-axis at } (0, 28).$$

Shape

The coefficient of $(x - 2)^2$ is negative, so draw a symmetric downward parabola about $x = 2$.



Solved reference diagram

Final answer

Downward parabola with turning point $(2, 52)$ and y -intercept $(0, 28)$.

Dependency check Sequenced immediately after WU027; the printed equation is repeated, so the crop is self-contained.

Question 11

4MA1-1H Q12 (a), (b)

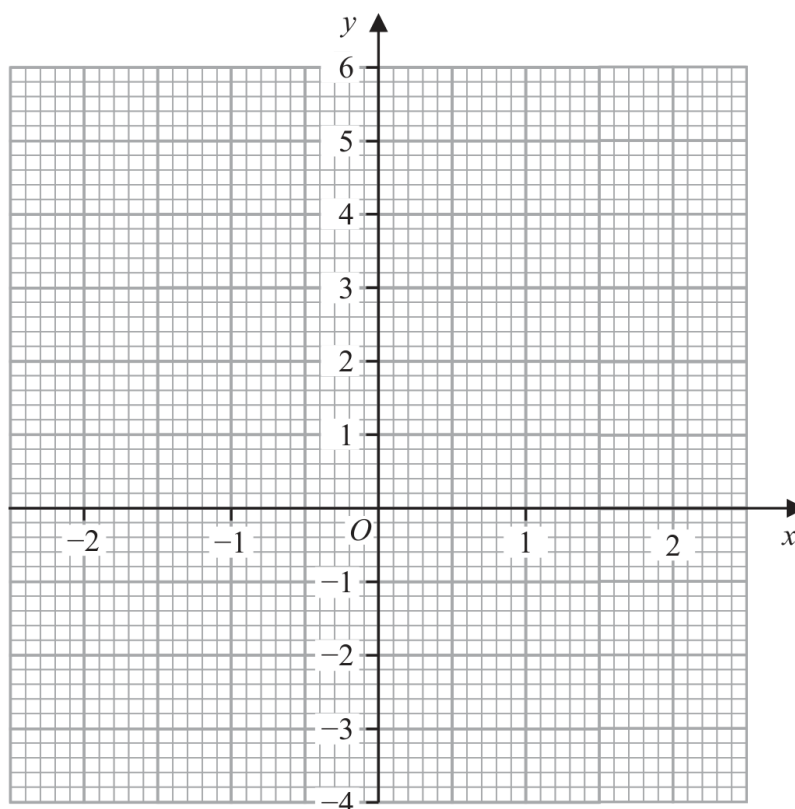
ORIGINAL QUESTION UNIT 1

4 marks

12 (a) Complete the table of values for $y = x^3 - 3x + 1$

x	-2	-1	0	1	2
y				-1	

(2)

(b) On the grid, draw the graph of $y = x^3 - 3x + 1$ for values of x from -2 to 2

(2)

Worked solution 11

Complete the cubic table / Draw the cubic

MODULAR UNIT 1

4 marks

(a) – Complete the cubic table

Substitute each x

$$y = x^3 - 3x + 1 \Rightarrow y(-2, -1, 0, 1, 2) = (-1, 3, 1, -1, 3).$$

Final answer

y values: -1, 3, 1, -1, 3

(b) – Draw the cubic

Plot the table points

$$(-2, -1), (-1, 3), (0, 1), (1, -1), (2, 3).$$

Join smoothly

Draw one smooth cubic curve through all five points for $-2 \leq x \leq 2$.

Final answerSmooth cubic through $(-2, -1), (-1, 3), (0, 1), (1, -1), (2, 3)$

Completed diagram 11

Complete the cubic table / Draw the cubic

MODULAR UNIT 1

4 marks

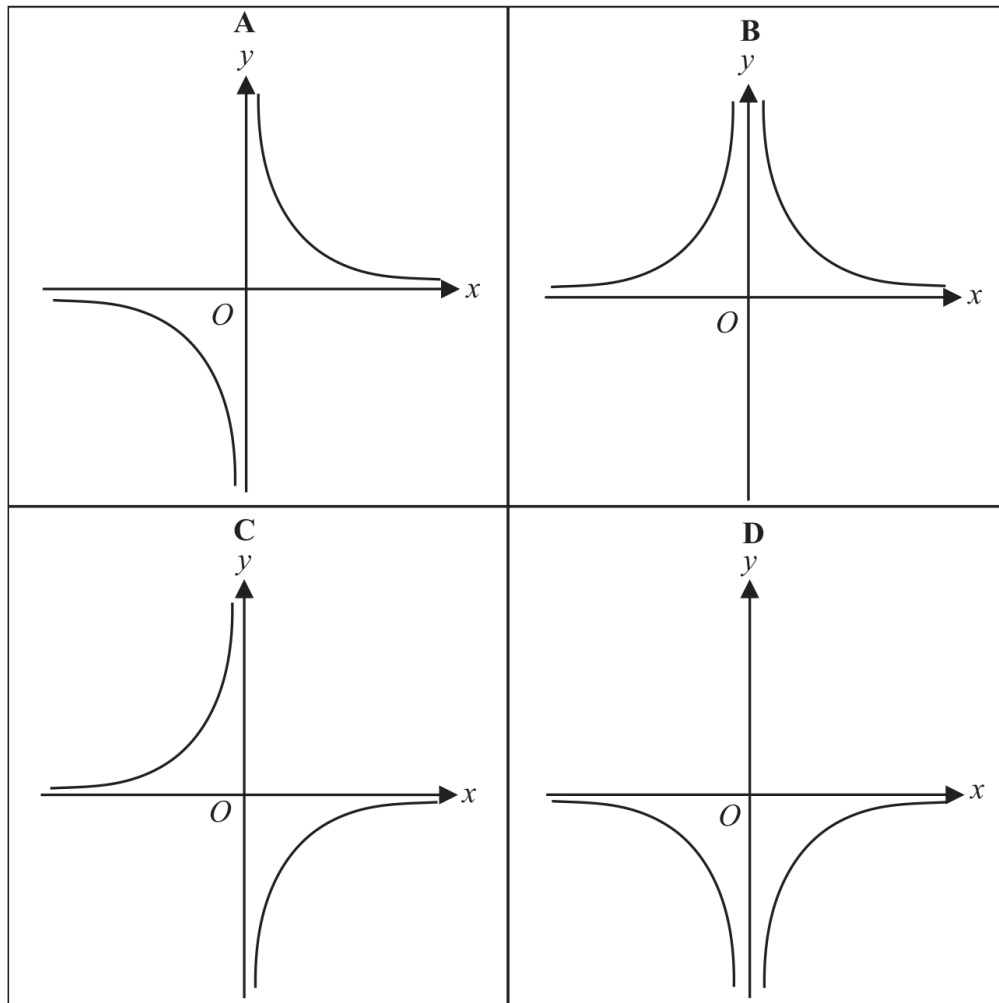
Question 12

4MA1-1H Q12 (c)

ORIGINAL QUESTION UNIT 1

1 marks

Here are four graphs.



(c) Write down the letter of the graph that could have the equation $y = -\frac{1}{x^2}$

.....
(1)

Worked solution 12Recognise $y = -1/x^2$

MODULAR UNIT 1

1 marks

(c) – Recognise $y = -1/x^2$

Use the sign and asymptotes

 $y = -1/x^2 < 0$ for $x \neq 0$, with branches in quadrants III and IV; this is graph D.**Final answer**

D

Question 15

4MA1-2H Q4

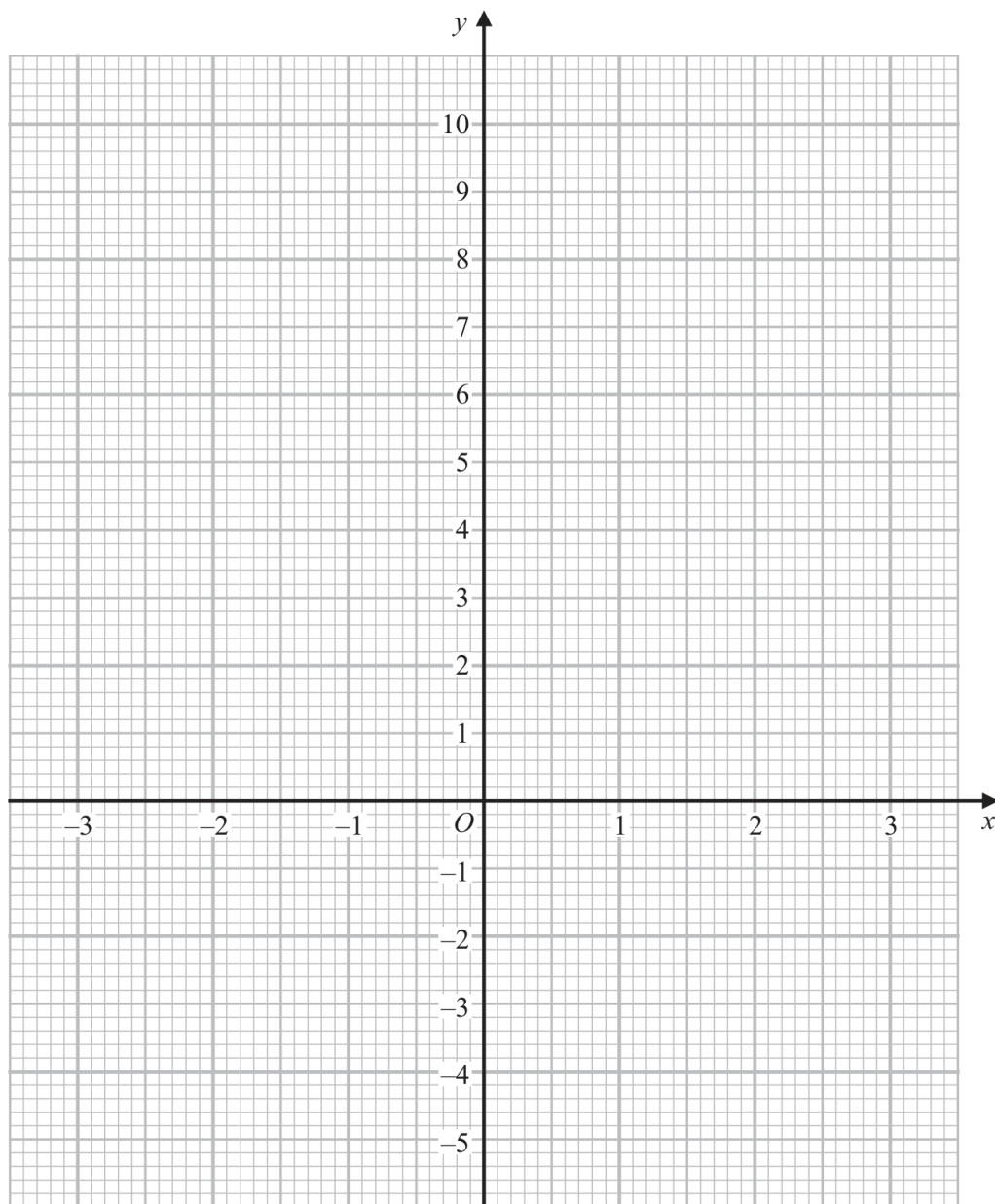
ORIGINAL QUESTION**4 marks**

- 4 (a) Complete the table of values for $y = x^2 - x - 4$

x	-3	-2	-1	0	1	2	3
y		2			-4		

(2)

- (b) On the grid below, draw the graph of $y = x^2 - x - 4$ for values of x from -3 to 3



(2)

Worked Solution 15

Complete and draw a quadratic graph

MODULAR UNIT 1

4 marks

Complete the table

$$y = x^2 - x - 4 \Rightarrow (8, 2, -2, -4, -4, -2, 2) \text{ for } x = -3, -2, -1, 0, 1, 2, 3.$$

Plot and join

Plot $(-3, 8), (-2, 2), (-1, -2), (0, -4), (1, -4), (2, -2), (3, 2)$ and join with a smooth quadratic curve.

Final answer

$$y = x^2 - x - 4; \quad y\text{-values } 8, 2, -2, -4, -4, -2, 2$$

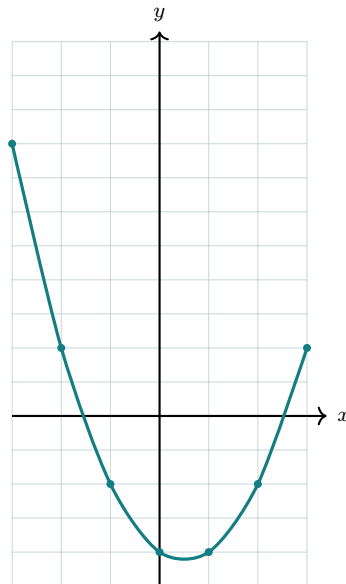
Completed diagram 15

Complete and draw a quadratic graph

MODULAR UNIT 1

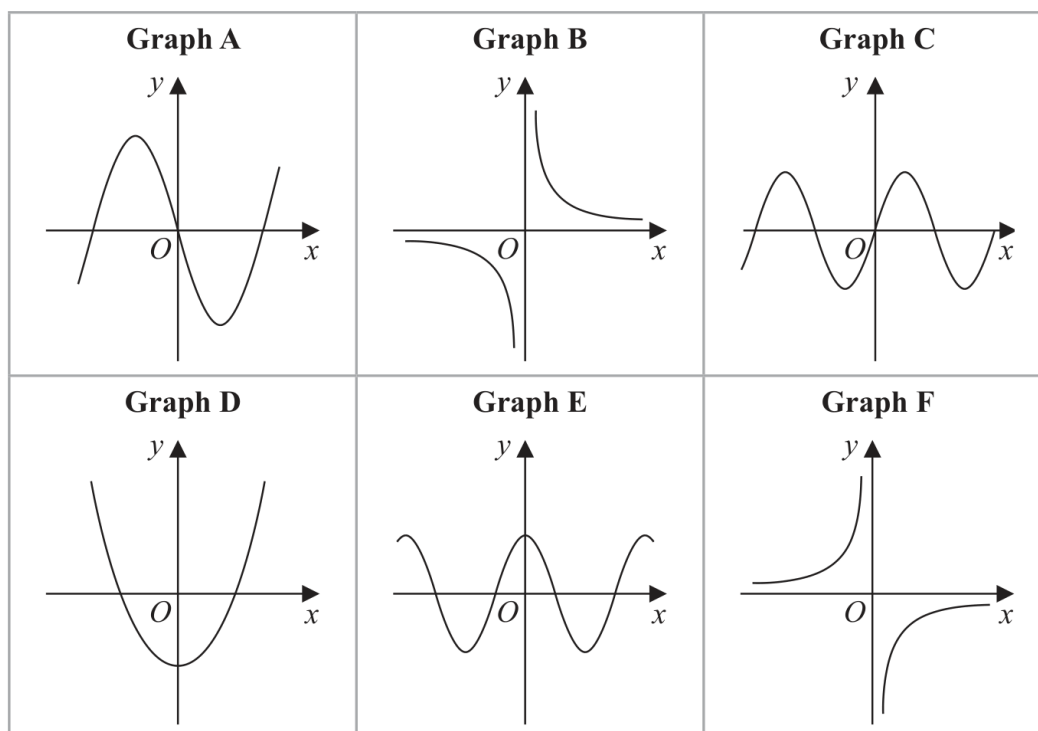
4 marks

Completed solution diagram: Completed quadratic graph



Question 23

4MA1-2H Q18

ORIGINAL QUESTION**3 marks****18** Here are 6 graphs.

Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \sin x$	
$y = -\frac{3}{x}$	
$y = 4x^3 - 5x$	

Worked Solution 23

Recognise function graph families

MODULAR UNIT 1

3 marks

Match the sine graph $y = \sin x$ passes through the origin and oscillates : Graph C.**Match the reciprocal graph** $y = -3/x$ has positive branch for $x < 0$ and negative branch for $x > 0$: Graph F.**Match the cubic** $y = 4x^3 - 5x$ is an odd cubic with negative central gradient : Graph A.**Final answer**

$$\sin x \rightarrow C, \quad -3/x \rightarrow F, \quad 4x^3 - 5x \rightarrow A$$

Original question 08

4MA1-1H Q11 M01, M02

MODULAR UNIT 1

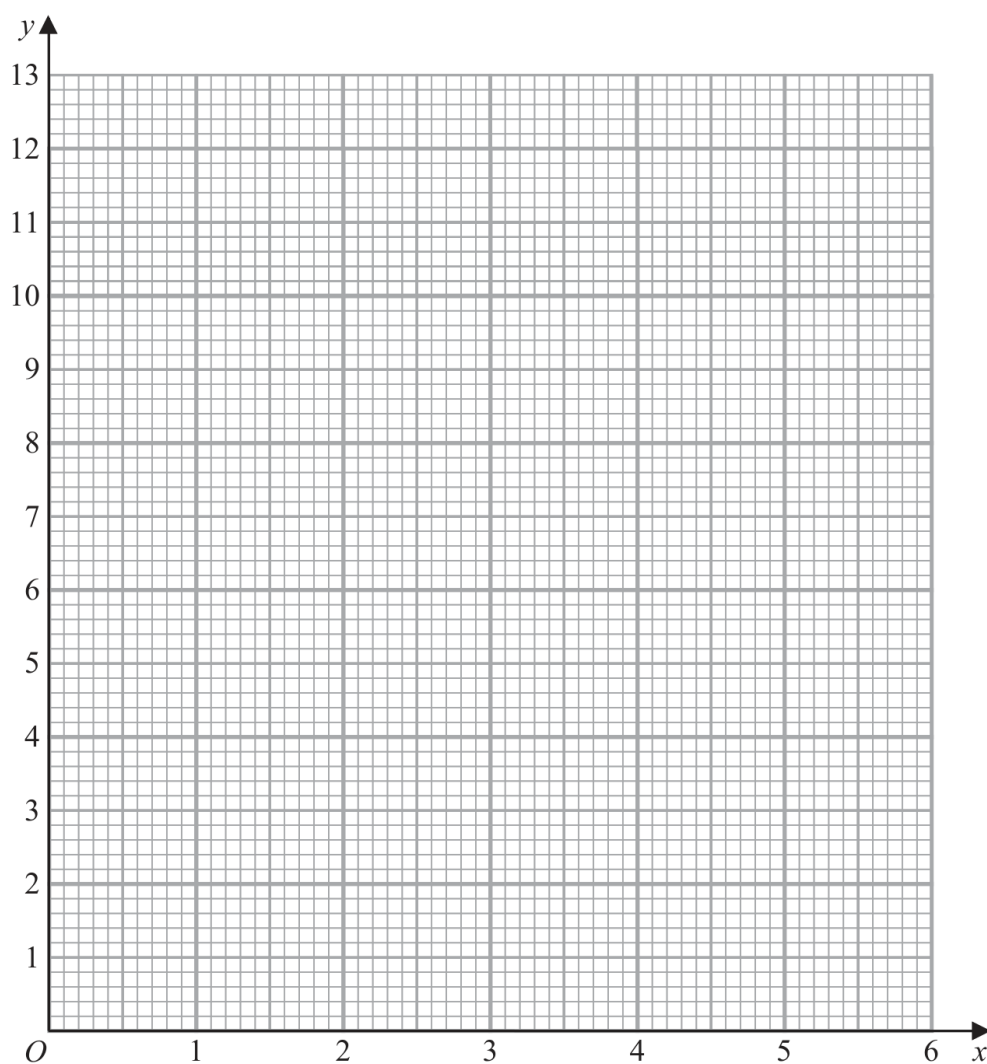
4 marks

- 11 (a) Complete the table of values for $y = 2\left(x + \frac{1}{x}\right)$

x	0.5	1	2	3	4	5	6
y			5	6.7			12.3

(2)

- (b) On the grid, draw the graph of $y = 2\left(x + \frac{1}{x}\right)$ for $0.5 \leq x \leq 6$



(2)

Worked solution 08

Complete a function table / Draw the function graph

MODULAR UNIT 1

4 marks

M01 – Complete a function table**Use the displayed formula**

$$y = 2\left(x + \frac{1}{x}\right).$$

Substitute the four missing x-values

$$y(0.5) = 5, \quad y(1) = 4, \quad y(4) = 8.5, \quad y(5) = 10.4.$$

Final answer

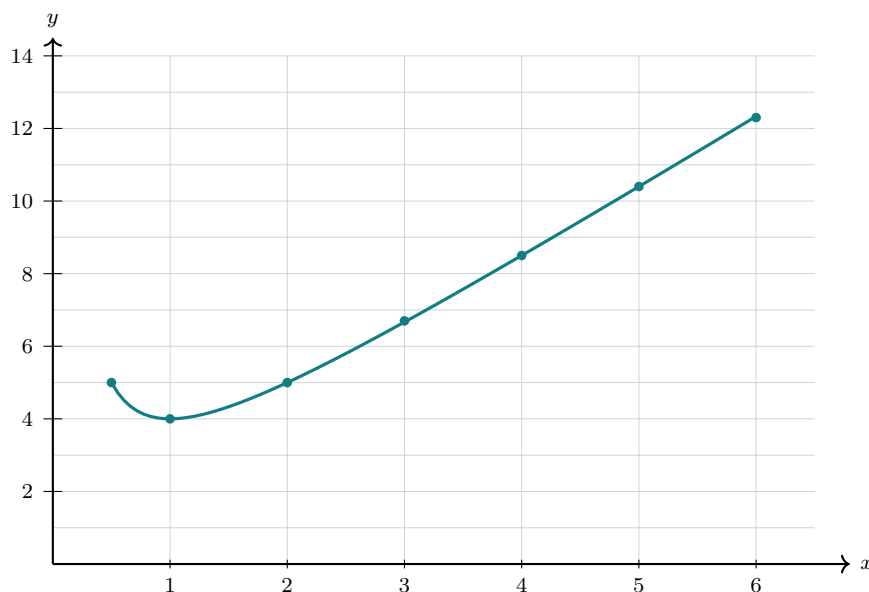
Missing values: 5, 4, 8.5, 10.4

M02 – Draw the function graph**Plot the table**Plot the seven table points for $0.5 \leq x \leq 6$.**Join smoothly**

Join the points with the correct smooth curve on the given grid.

Final answer

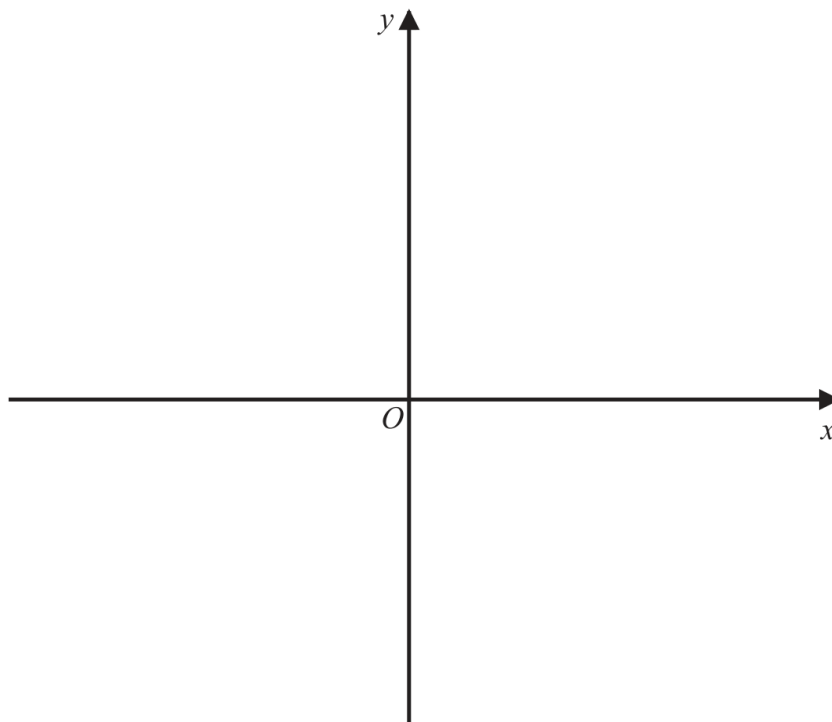
The correct smooth curve through the table points

Completed solution diagram: Smooth curve through the table points

20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.
On your sketch show clearly, in terms of a ,

- (i) the coordinates of any points of intersection of **C** with the coordinate axes,
- (ii) the coordinates of the turning point.



Worked solution

January 2020 | Question 20 | 4 marks

Family: Sketch key features without a full table • Variant: Feature-led polynomial sketch

Method / working

1. Set $y=0$: $4(x-1)^2=a$, so $x=1\pm\sqrt{a}/2$.

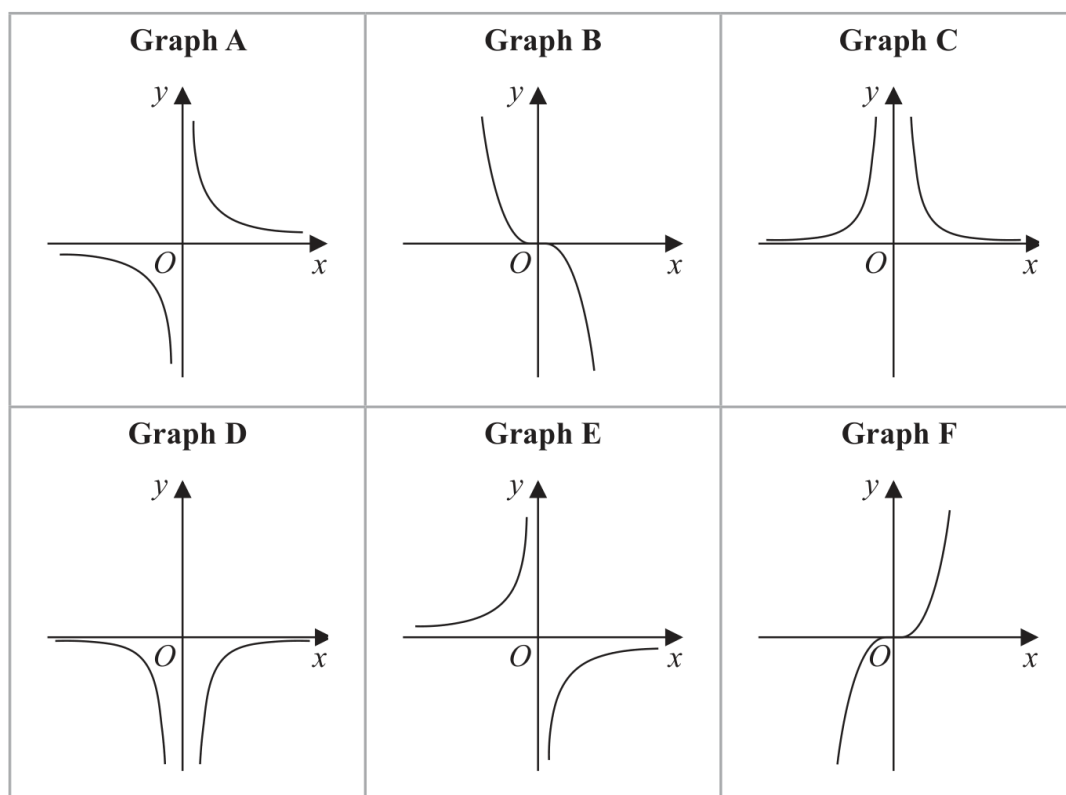
2. At $x=0$, $y=4-a$.

3. The vertex form shows turning point $(1,-a)$.

4. Sketch an upward parabola and label these points.

Final answer: Intercepts $(1\pm\sqrt{a}/2,0)$; y-intercept $(0,4-a)$; turning point $(1,-a)$

15 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \frac{2}{x^2}$	
$y = -\frac{1}{2}x^3$	
$y = -\frac{5}{x}$	

Worked solution

January 2020 | Question 15 | 3 marks

Family: Recognise a function graph family | Variant: Recognise polynomial graph shape

METHOD / WORKING

1. $y = -2/x^2$ is negative and y – axis symmetric : choose graph C.2. $y = -(1/2)x^3$ is a decreasing cubic through the origin : choose graph B.3. $y = -5/x$ is a negative reciprocal graph: choose graph E.

FINAL ANSWER

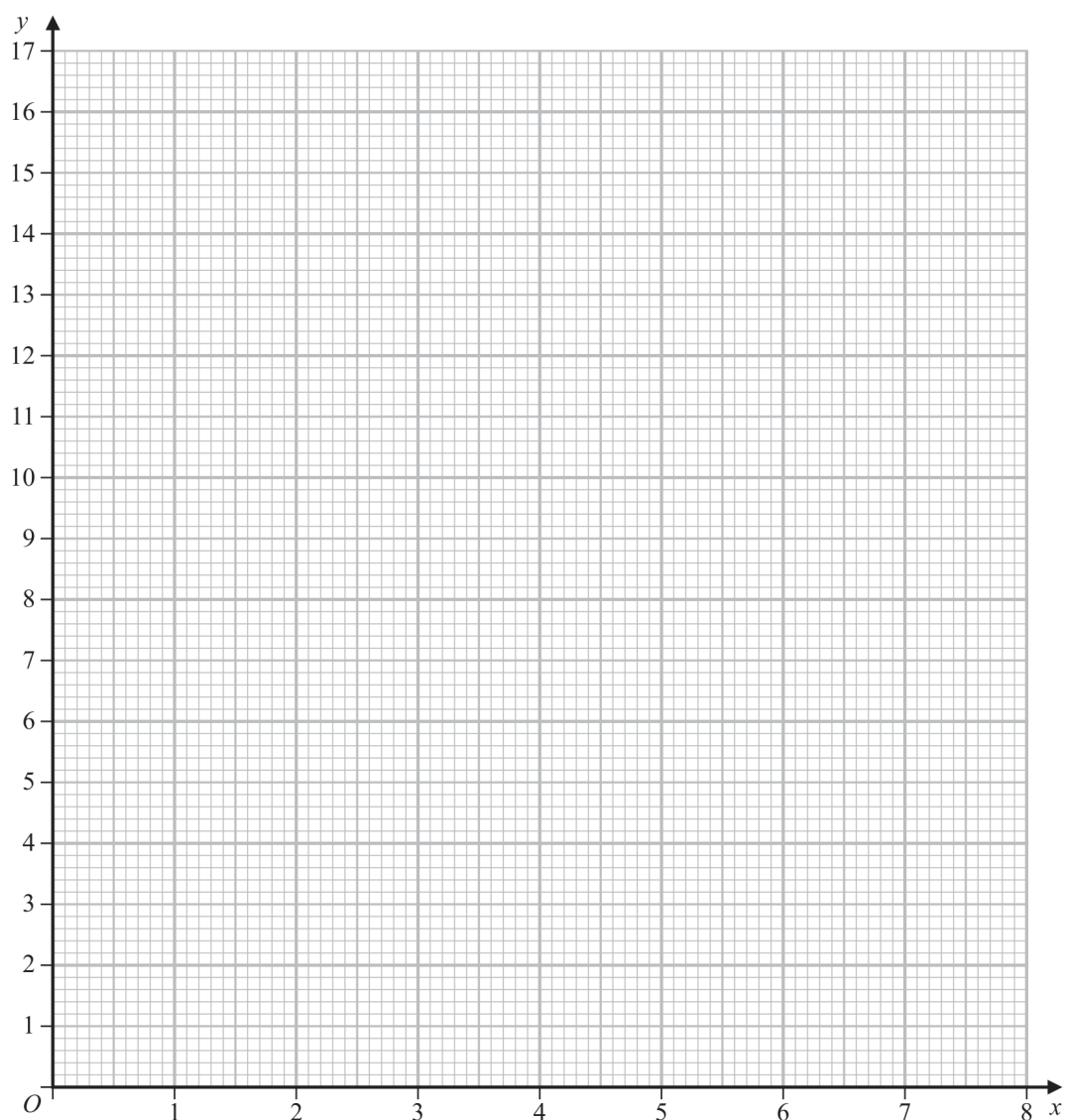
C, B, E

15 (a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

x	0.25	0.5	1	2	4	8
y	16.25					8.5

(2)

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leq x \leq 8$



(2)

Worked solution

June 2021 | Question 15 | 2 marks

Family: Generate a table and draw a nonlinear graph | Variant: Plot a reciprocal or mixed rational curve

METHOD / WORKING

1. Use $y = (x^2 + 4)/x = x + 4/x$ for $x = .5, 1, 2$ and 4 .

2. The missing values are 8.5, 5, 4 and 5 respectively.

FINAL ANSWER

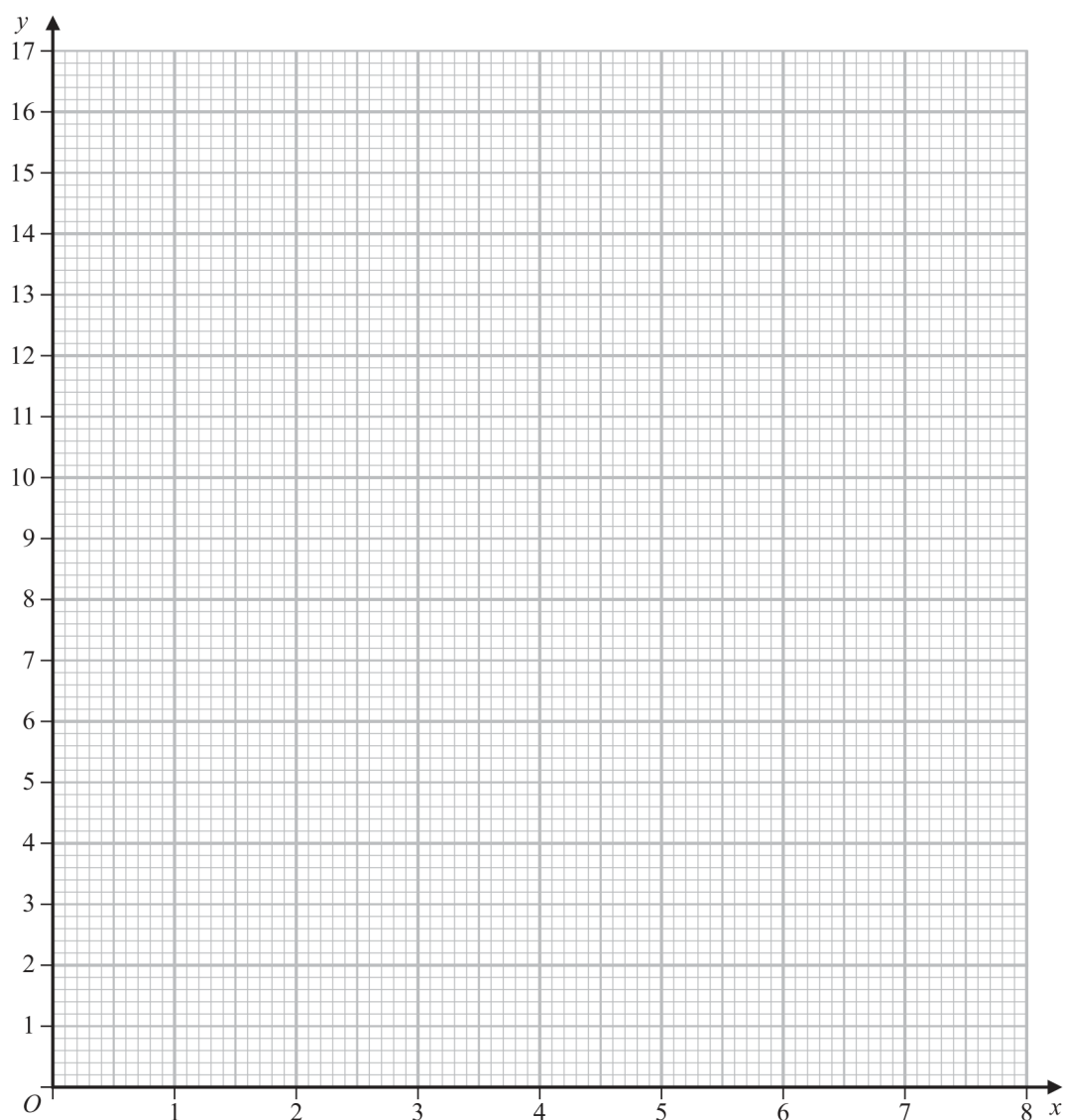
8.5, 5, 4, 5

15 (a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

x	0.25	0.5	1	2	4	8
y	16.25					8.5

(2)

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leq x \leq 8$



(2)

Worked solution

June 2021 | Question 15 | 2 marks

Family: Generate a table and draw a nonlinear graph • Variant: Plot a reciprocal or mixed rational curve

Method / working

1. Plot all six points from the completed table accurately on the supplied grid.
2. Join them with a smooth curve for $0.25 \leq x \leq 8$.

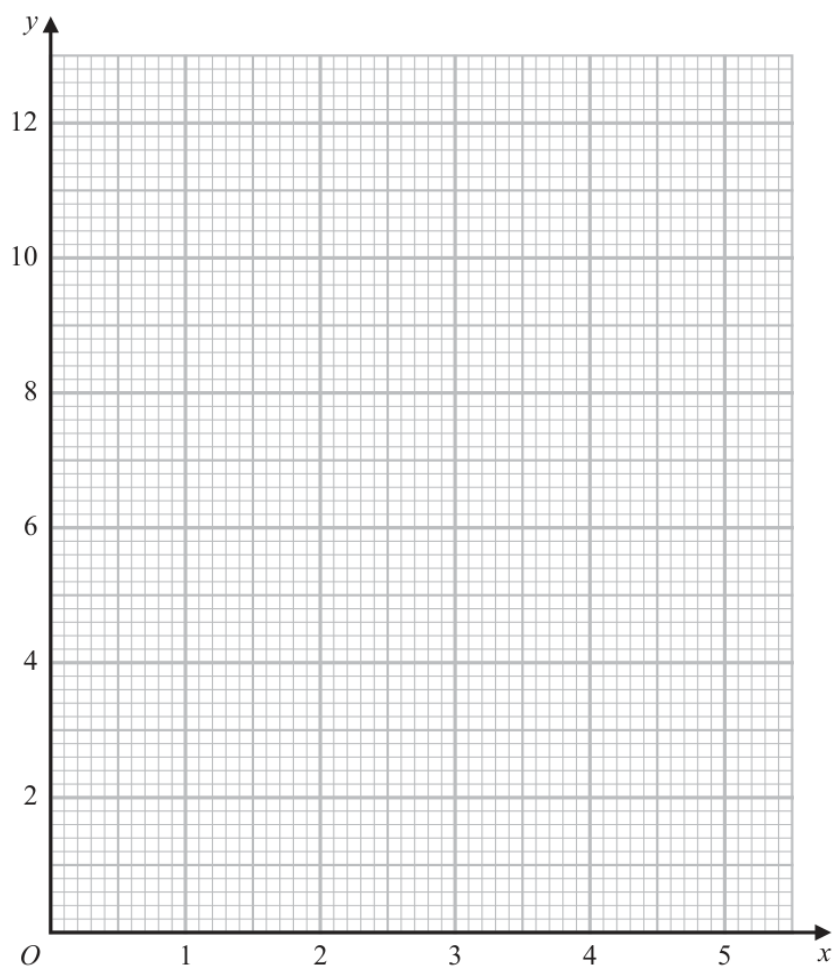
Final answer: Smooth graph through the six table points

14 (a) Complete the table of values for $y = \frac{2}{x}\left(5 - \frac{1}{x}\right)$

x	0.5	1	2	3	4	5
y		8		3.1	2.4	1.9

(1)

(b) On the grid, draw the graph of $y = \frac{2}{x}\left(5 - \frac{1}{x}\right)$ for $0.5 \leq x \leq 5$



(2)

(Total for Question 14 is 3 marks)

Answer reference | January 2023 | Question 14

Family: Generate a table and draw a nonlinear graph • Variant: Plot a cubic from a table

14	(a)		12 and 4.5	1	B1 allow $\frac{9}{2}$ oe May be awarded if plotted correctly on the graph
	(b)		Correct graph	2	M1 ft for at least 5 points plotted correctly ($\pm$ half square)
		Correct answer scores full marks (unless from obvious incorrect working)			A1 for correct curve between $x = 0.5$ and $x = 5$ (clear intention to go through all the points and which must be curved) Note: If a fully correct graph is shown, but an incomplete table is shown in (a), then award the marks for (a)
					Total 3 marks

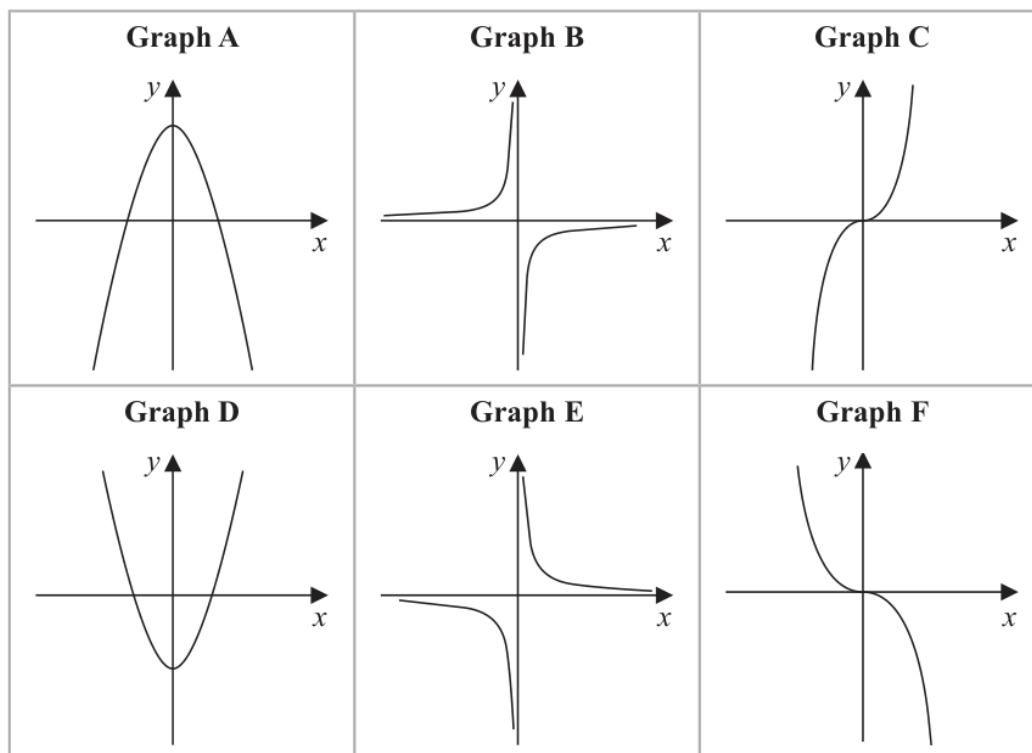
Worked solution

January 2023 | Question 14 | 3 marks

Family: Generate a table and draw a nonlinear graph • Variant: Plot a cubic from a table

Official final mark-scheme pages are embedded immediately after this question.

11 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = -\frac{2}{x}$	
$y = 5 - x^2$	
$y = -2x^3$	

(Total for Question 11 is 3 marks)

Answer reference | January 2023 | Question 11

Family: Generate a table and draw a nonlinear graph • Variant: Plot a cubic from a table

Question	Working	Answer	Mark	Notes
11		B	3	B1
		A		B1
		F		B1
				Total 3 marks

Worked solution

January 2023 | Question 11 | 3 marks

Family: Generate a table and draw a nonlinear graph • Variant: Plot a cubic from a table

Official final mark-scheme pages are embedded immediately after this question.

Original question 15

4MA1-1H Q23 M01

MODULAR UNIT 2

5 marks

- 23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)

The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k

Show clear algebraic working.

Original question 15 (continued)

4MA1-1H Q23 M01

MODULAR UNIT 2

5 marks

 $k =$

Worked solution 15Use the chord separation to find k

MODULAR UNIT 2

5 marks

M01 – Use the chord separation to find k **Use the equal y -values**

$$p^2 - 8p - 9 = k \text{ and } q^2 - 8q - 9 = k.$$

Use symmetryThe axis is $x=4$, so $(p+q)/2=4$; with $p-q=14$, $p=11$, $q=-3$.**Substitute**

$$k = 11^2 - 8(11) - 9 = 24.$$

Final answer

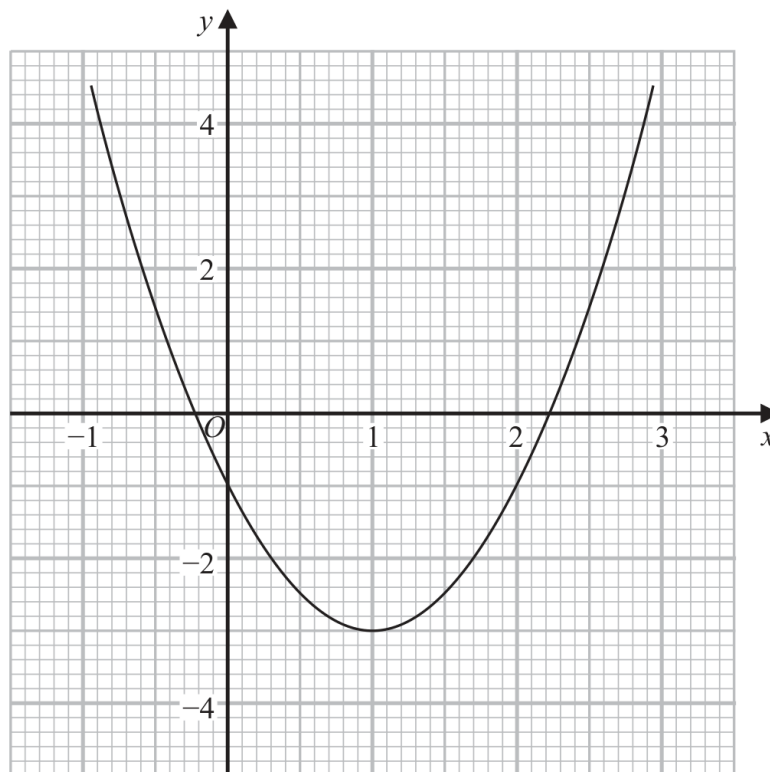
$$k = 24$$

Original question 32

4MA1-2H Q21 Q21(a)

ORIGINAL QUESTION**2 marks**

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



- (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

.....
(2)

Worked solution 32

Chapter: Graphs of Functions

MODULAR UNIT 2

2 marks

Q21(a) – Chapter: Graphs of Functions

Family: Solve using graph roots or intersections • Variant: Estimate roots from axis crossings

MethodThe x-intercepts of $y=2x^2-4x-1$ are approximately $x=-0.2$ and $x=2.2$.*Why: Read where the curve crosses the x-axis.***Method**

Both values are given to one decimal place.

*Why: Round the graph readings as requested.***Final answer****roots:** -0.2 and 2.2

Original question 33

4MA1-2H Q21 Q21(b)

ORIGINAL QUESTION**3 marks**

- (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

.....
(3)

Worked solution 33

Chapter: Graphs of Functions

MODULAR UNIT 2

3 marks

Q21(b) – Chapter: Graphs of Functions

Family: Solve using graph roots or intersections • Variant: Draw a line to solve a nonlinear equation

MethodFor $y=x^2-x-1$, use $y=-2x+1$ as the straight line.*Why: Rearrange $x^2-x-1=0$ to $x^2-x=1$ and compare with the displayed quadratic.***Method**Draw $y=-2x+1$ through $(0,1)$ and $(-1,3)$.*Why: Use two points to draw the line on the supplied grid.***Method**The intersections give $x \approx -0.6$ and $x \approx 1.6$.*Why: Read the two x -coordinates to one decimal place.***Final answer**roots: -0.6 and 1.6

Completed diagram 33

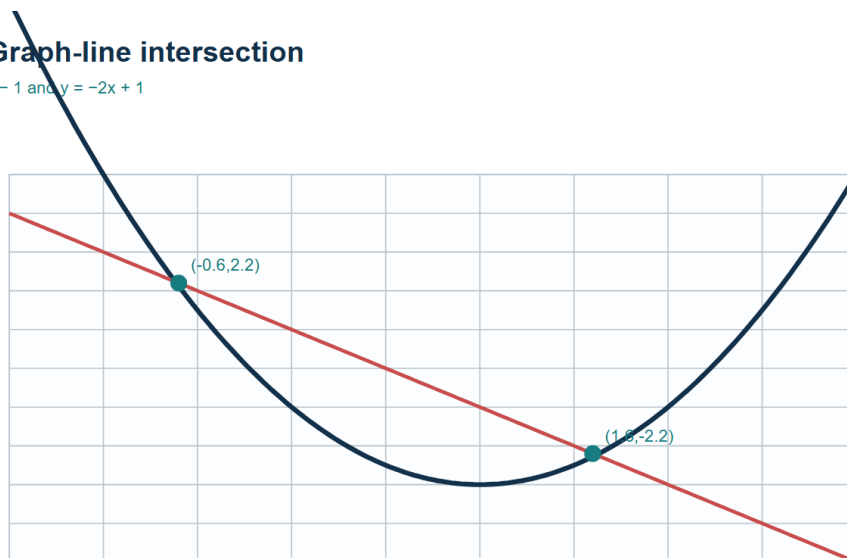
Chapter: Graphs of Functions

MODULAR UNIT 2

3 marks

Q21 • Graph-line intersection

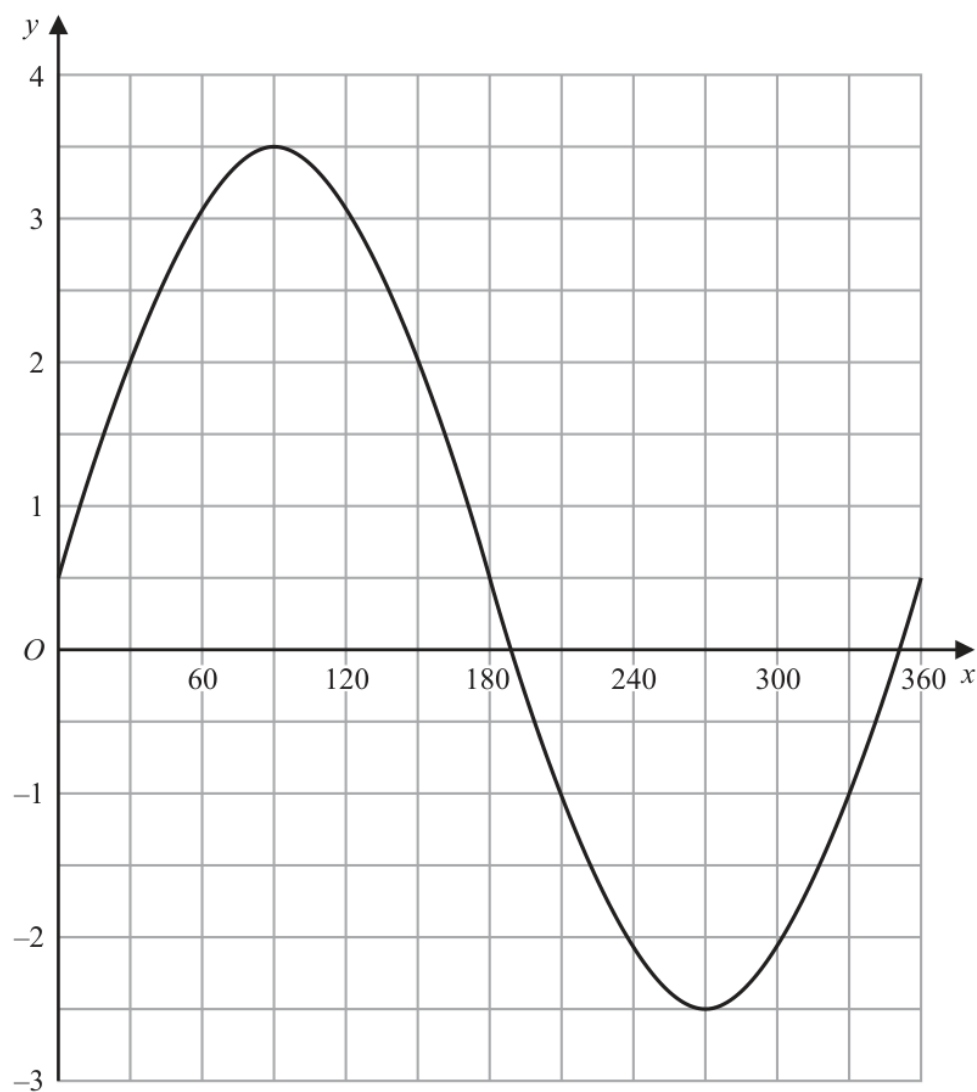
$$y = 2x^2 - 4x - 1 \text{ and } y = -2x + 1$$



x-intersections: -0.6 and 1.6

Elite IGCSE Mathematics • Dr Eslam Ahmed • eliteigcse.com • WhatsApp +20 112 000 9622

24 The graph of $y = a \sin x^\circ + b$ is drawn on the grid.



Find the value of a and the value of b

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(Total for Question 24 is 2 marks)

Answer reference | January 2023 | Question 24

Family: Solve using graph roots or intersections • Variant: Draw a line to solve a nonlinear equation

Q	working	Answer	Mark	Notes
24		3	2	B1
		0.5		B1 oe
				Total 2 marks

Worked solution

January 2023 | Question 24 | 2 marks

Family: Solve using graph roots or intersections • Variant: Draw a line to solve a nonlinear equation

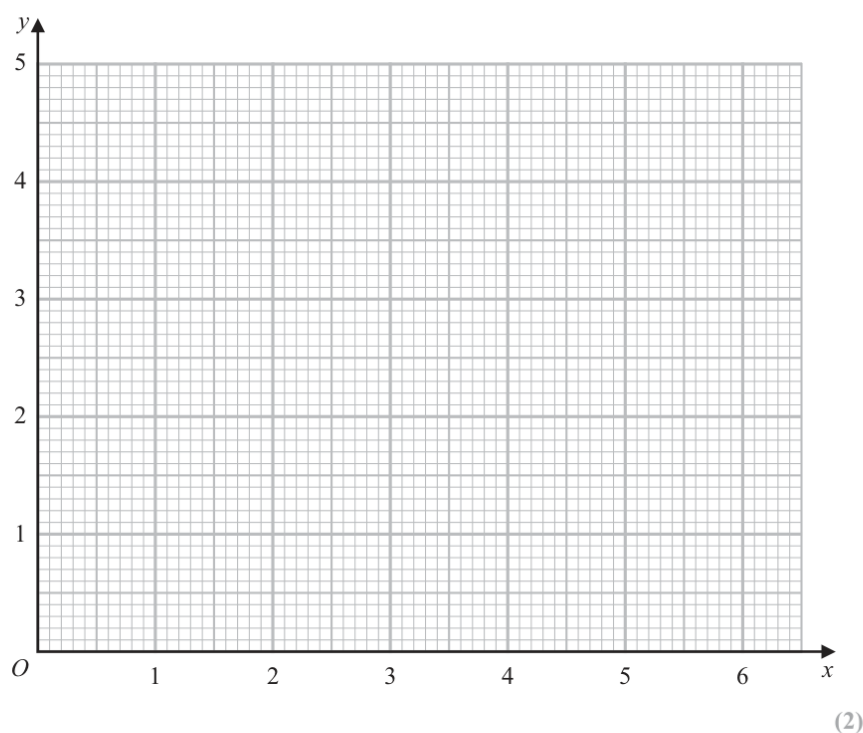
Official final mark-scheme pages are embedded immediately after this question.

12 (a) Complete the table of values for $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$

x	0.5	1	2	3	4	5	6
y	4.25			2.17	2.5	2.9	3.33

(1)

(b) Draw the graph of $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$ for $0.5 \leq x \leq 6$



- (c) By drawing a suitable line on the grid, find estimates for the solutions of the equation $x + \frac{4}{x} = 6$

Give your answers correct to one decimal place.

.....
(2)

(Total for Question 12 is 5 marks)

Worked solution

November 2025 | Question 12 | 5 marks

Family: Solve using graph roots or intersections • Variant: Draw a line to solve a nonlinear equation

Solution

$$y = \frac{1}{2}\left(x + \frac{4}{x}\right)$$

The completed table is:

$\begin{array}{c|cccccc} \end{array}$

$x \quad 0.5 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad \backslash$

y

$4.25 \quad 2.5 \quad 2 \quad 2.17 \quad 2.5 \quad 2.9 \quad 3.33$

$\end{array}$

For

$$x + \frac{4}{x} = 6$$

we need

$$\frac{1}{2}\left(x + \frac{4}{x}\right) = 3$$

So draw the horizontal line $y=3$ on the graph.

The intersections give approximately

$$x=0.8 \text{ quad and quad } x=5.2$$

Answer: Table missing values 2.5, 2. Estimates 0.8 and 5.2.